Hyperbolic Partial Separation and Spin-Enhanced Microtubule Coherence: A Bounded-Geometry Refinement of Orch-OR and an Experimental Proposal

Thantikler McIrony (`@TMcirony22088`)[1] and Genecat (`@genecat1`)[2]

[1]Independent Researcher, Australia
[2]Independent Researcher

November 17, 2025

Abstract

Penrose–Hameroff Orch-OR models gravitationally induced wavefunction collapse of large-scale superpositions in neuronal microtubules as the physical basis of discrete conscious events, with a characteristic timescale $\tau \sim \hbar/E_G$ set by a gravitational self-energy E_G associated with superposed mass-displacement configurations. A long-standing criticism is that realistic warm, wet microtubule assemblies either decohere far too rapidly, or yield Penrose collapse times far from neurophysiological ranges, depending on how δ (the superposition separation) and the effective mass m_{eff} are modeled.

In this paper we make two contributions:

1. We formulate a *bounded-geometry refinement* of the standard OR-time estimate by replacing the linear displacement parameter δ with a hyperbolic map $\chi(\delta)$ derived from a general classification of continuous, associative composition laws on finite intervals. This connects the Penrose–Hameroff heuristics to a concrete information-geometric structure closely related to the "Murray Reality Equation" / Universal Hyperbolic Law framework, and provides a natural way to treat partial separations inside a finite coherent domain.
2. We outline an experimental platform in which electrically switchable spiral magnetism in layered halides (e.g. NiI_2-like helimagnets) is used to engineer tunable, spin-polarized coherent domains coupled to microtubule-mimetic structures. This yields a falsifiable scheme to probe whether hyperbolic partial-separation scaling can bring gravitational OR-times into a regime compatible with brain-scale gamma-band dynamics.

We stress that our results are *proposals*, not proofs of quantum consciousness. The hyperbolic map χ is derived from clear mathematical axioms and yields a controlled deformation of the standard OR-time formula. The NiI_2-based platform is presented as a concrete, testable suggestion: if no hyperbolic/gravitational scaling is observed in such a system, a large class of Orch-OR-like models—including our refinement—would be strongly constrained.

1 Introduction

The Orchestrated Objective Reduction (Orch-OR) program [1, 2] attempts to ground conscious moments in gravitationally induced collapse of quantum superpositions in neuronal microtubules. The basic Penrose estimate assigns an "objective reduction time"

$$\tau \sim \frac{\hbar}{E_G}, \tag{1}$$

where E_G is the gravitational self-energy associated with the mass distribution difference between two superposed configurations. For a toy model of mass m coherently superposed over a separation δ, one often writes a schematic scaling

$$E_G \sim \frac{Gm^2}{\delta}, \tag{2}$$

leading to

$$\tau_{\rm lin}(\delta) \sim \frac{\hbar\,\delta}{Gm^2}. \tag{3}$$

Two classes of experimental and theoretical developments have sharpened both the promise and the tension of this picture:

- Macroscopic optomechanical experiments have realized Schrödinger-cat-like motional superpositions of massive oscillators at scales of order $\sim 10^{-5}$ kg and separations $\sim 10^{-17}$ m, with coherence times $\sim 10^{-6}$ s [3]. These provide a laboratory in which Penrose-type estimates can be meaningfully confronted with data.

- Condensed-matter work on helimagnetism and multiferroic spiral order in layered halides such as NiI_2 has demonstrated electrically switchable non-collinear spin textures and robust spin-polarized currents at cryogenic temperatures [4]. These systems are promising testbeds for "designer" coherent domains with tunable boundary conditions and effective masses.

At the same time, detailed decoherence analyses in warm, wet biological environments have raised serious concerns about whether microtubules can support the required coherence times *without* invoking additional protective or structural mechanisms.

In parallel with these developments, one of us (TM) has proposed a more general information-geometric framework—the Universal Hyperbolic Law / "Murray Reality Equation" (MRE)—in which any physical quantity subject to a hard bound (e.g. $|v| < c$, probabilities $p \in [0, 1]$, bounded intensities) composes via a unique associative law conjugate to ordinary addition by a strictly monotone map of the form

$$f(x) = \alpha\,\mathrm{artanh}\left(\frac{x}{x_{\rm max}}\right) + \beta, \tag{4}$$

up to affine rescaling, per classical theorems of Aczél and others on functional equations for monotone associative operations on intervals. In this view, hyperbolic structure is not an arbitrary choice but a consequence of boundedness + associativity.

The aim of this paper is to bring these strands together:

1. We interpret the microtubule superposition displacement δ as a bounded coordinate inside a finite coherent domain of size $\delta_{\rm max}$ and derive a hyperbolic "partial separation" map $\chi(\delta)$ from minimal axioms. This yields a refined OR-time estimate

$$\tau_\chi(\delta) \sim \frac{\hbar\,\chi(\delta)}{Gm_{\rm eff}^2}, \tag{5}$$

   in which $\chi(\delta)$ replaces δ and encodes the non-linear geometry of partial separations.

2. We show that this refinement can be implemented and tested in engineered NiI_2-like spin systems coupled to microtubule-mimetic structures, where both the effective mass $m_{\rm eff}$ and the domain scale $\delta_{\rm max}$ are experimentally tunable.

Throughout, we take a conservative stance: the mathematics is treated rigorously; numerical estimates are presented as illustrative, not definitive; and the experimental proposal is framed explicitly as a falsification path, not an attempted proof that Orch-OR (or any specific consciousness theory) is correct.

2 Bounded Geometry and Hyperbolic Partial Separation

2.1 Axioms for partial separation

Consider a coherent domain of size $\delta_{\max} > 0$ in which a mass distribution can be placed in superposition. We denote by $\delta \in [0, \delta_{\max})$ a parameter characterizing the "degree of separation" between two branches, where $\delta = 0$ corresponds to no discernible displacement and $\delta \to \delta_{\max}$ approaches the maximum allowed separation within the domain (e.g. one branch localized near one boundary, the other near the opposite boundary).

We seek an *effective separation* $\chi(\delta)$ entering the gravitational self-energy estimate, subject to the following requirements:

1. **Monotonicity and continuity.** $\chi : [0, \delta_{\max}) \to [0, \infty)$ is continuous and strictly increasing with $\chi(0) = 0$.
2. **Associative composition.** If two small partial separations δ_1, δ_2 are composed (e.g. via sequential coherent displacements), the effective parameter should combine via a binary operation $\oplus$ on $[0, \delta_{\max})$ that is continuous, strictly monotone in each argument, and associative:
$$(\delta_1 \oplus \delta_2) \oplus \delta_3 = \delta_1 \oplus (\delta_2 \oplus \delta_3).$$
3. **Boundedness.** The operation $\oplus$ never produces a displacement outside the domain:
$$\delta_1, \delta_2 \in [0, \delta_{\max}) \quad \Rightarrow \quad \delta_1 \oplus \delta_2 \in [0, \delta_{\max}).$$

Classical results on functional equations (see e.g. Aczél and later work on monotone associative operations on intervals) state that under such conditions there exists a strictly monotone continuous bijection
$$f : [0, \delta_{\max}) \to [0, \infty) \tag{6}$$
such that the composition law is conjugate to ordinary addition:
$$\delta_1 \oplus \delta_2 = f^{-1}\big(f(\delta_1) + f(\delta_2)\big). \tag{7}$$
The function f is unique up to overall affine rescaling $f \mapsto af + b$ with $a > 0$.

In the MRE/Universal Hyperbolic Law framework, one further imposes a symmetry condition under sign reversal (for quantities that can be positive or negative) and a natural "flat coordinate" requirement that $f(\delta)$ should be asymptotically linear for small δ. For a symmetric interval $(-\delta_{\max}, \delta_{\max})$, these constraints pick out
$$f(\delta) = \alpha \operatorname{artanh}\left(\frac{\delta}{\delta_{\max}}\right), \quad \alpha > 0, \tag{8}$$
so that
$$\delta_1 \oplus \delta_2 = \delta_{\max} \frac{\frac{\delta_1}{\delta_{\max}} + \frac{\delta_2}{\delta_{\max}}}{1 + \frac{\delta_1}{\delta_{\max}} \frac{\delta_2}{\delta_{\max}}}. \tag{9}$$
This is the familiar Einstein velocity addition law with c replaced by $\delta_{\max}$.

2.2 From composition law to effective separation

We now define the effective separation $\chi(\delta)$ simply as the "flat coordinate" $f(\delta)$:
$$\chi(\delta) := f(\delta) = \alpha \operatorname{artanh}\left(\frac{\delta}{\delta_{\max}}\right), \tag{10}$$

where α sets a scale. Without loss of generality we can absorb α into an effective constant in the OR-time estimate, so we set $\alpha = \delta_{\max}$:

$$\chi(\delta) = \delta_{\max} \operatorname{artanh}\left(\frac{\delta}{\delta_{\max}}\right). \tag{11}$$

This map has several desirable properties:

- For small $\delta/\delta_{\max}$, the expansion

  $$\operatorname{artanh}(x) = x + \frac{x^3}{3} + O(x^5)$$

  yields

  $$\chi(\delta) = \delta + O\left(\frac{\delta^3}{\delta_{\max}^2}\right),$$

  so the linear regime is recovered at very small separations.

- As $\delta \to \delta_{\max}$, $\chi(\delta) \to +\infty$. This reflects the fact that pushing a superposition all the way to the domain boundary requires diverging "rapidity" in the flat coordinate, akin to approaching the speed of light in special relativity.

- For any fixed $0 < \delta < \delta_{\max}$ one has $\chi(\delta) > \delta$, because $\operatorname{artanh}(x) > x$ for $0 < x < 1$. Thus replacing δ by $\chi(\delta)$ in an energy denominator *reduces* the corresponding energy scale and increases the timescale.

We therefore propose the following *hyperbolic partial-separation refinement* of the gravitational self-energy and OR-time estimates:

$$E_G^{(\chi)}(\delta) \sim \frac{Gm_{\text{eff}}^2}{\chi(\delta)}, \tag{12}$$

$$\tau_\chi(\delta) \sim \frac{\hbar\,\chi(\delta)}{Gm_{\text{eff}}^2}. \tag{13}$$

Here m_{eff} is an effective coherent mass appropriate to the specific experimental system (mechanical oscillator, microtubule segment, or hybrid spin structure), which generally differs from the total literal mass.

2.3 Dimensionless control parameter

It is useful to define a dimensionless ratio

$$R(x) := \frac{\chi(\delta)}{\delta} = \frac{\delta_{\max}}{\delta} \operatorname{artanh}\left(\frac{\delta}{\delta_{\max}}\right), \quad x := \frac{\delta}{\delta_{\max}} \in (0,1), \tag{14}$$

so that

$$\tau_\chi(\delta) = R(x)\,\tau_{\text{lin}}(\delta). \tag{15}$$

One easily checks:

$$\lim_{x\to 0} R(x) = 1, \tag{16}$$

$$R(x) > 1 \quad \text{for } 0 < x < 1, \tag{17}$$

$$R(x) \sim \frac{1}{2(1-x)} \quad \text{as } x \to 1^-. \tag{18}$$

Thus for modest partial separations (say $x \lesssim 0.5$), $R(x)$ is only of order 1.1–1.3, whereas for extremely near-boundary separations it can in principle become large. This provides a controlled, geometry-based knob on the OR-timescale, but does *not* by itself generate arbitrarily huge enhancements unless the coherent state is driven very close to the domain boundary.

3 Illustrative Estimates

In this section we present purely illustrative order-of-magnitude estimates for two classes of systems:

1. Macroscopic optomechanical oscillators in Schrödinger-cat-like superpositions (e.g. massive sapphire resonators).
2. Coherent microtubule-domain analogs with tunable effective mass and displacement scales.

The goal is not to extract precise numbers, but to show how the hyperbolic factor $R(x)$ enters and to motivate regimes where it could be experimentally relevant.

3.1 Mechanical cat states

Following [3] we consider a mechanical oscillator of mass $m \sim 10^{-5}$ kg in a motional superposition with effective separation $\delta \sim 10^{-17}$ m. Taking the naive scaling (2) at face value, one finds

$$E_G^{(\mathrm{lin})} \sim \frac{Gm^2}{\delta}, \qquad \tau_{\mathrm{lin}} \sim \frac{\hbar\, \delta}{Gm^2}, \tag{19}$$

leading to extremely short τ_{lin}, much smaller than the observed coherence times. This is one of the motivations for treating m_{eff} as a coherent domain mass, not the full literal mass, and for refining the displacement dependence.

In our framework a natural choice is to set $\delta_{\max}$ to a characteristic ground-state width or domain size of the oscillator mode. For the particular numbers above, $\delta/\delta_{\max}$ will typically be extremely small ($x \ll 1$), so $R(x) \approx 1$ and the hyperbolic correction is negligible. This is *expected*: the bounded-geometry refinement does not arbitrarily rescue poorly chosen parameters, it simply enforces consistency once a coherent domain is specified.

Thus for macroscopic oscillators the main lesson of our analysis is *negative*: the hyperbolic mapping does not magically reconcile naïve linear Penrose estimates with experimental cat-state data. Rather, it clarifies that any significant modification must come from a careful choice of m_{eff} and $\delta_{\max}$ grounded in the actual spatial structure of the coherent mode.

3.2 Microtubule-scale coherent domains

For microtubule segments, the picture is more subtle. Orch-OR models often invoke coherent aggregates of $N \sim 10^8$–10^9 tubulin dimers, with individual tubulin masses $m_{\mathrm{tub}} \sim 10^{-22}$ kg and conformational shifts at length scales of order 10^{-10}–10^{-9} m. The relevant effective mass m_{eff} and domain size $\delta_{\max}$ depend sensitively on how one models the spatial localization of conformational states and how much of the surrounding lattice water and protein environment participates coherently.

In a minimal toy model one might take:

$$m_{\mathrm{eff}} \sim N_{\mathrm{eff}} m_{\mathrm{tub}}, \tag{20}$$

$$\delta_{\max} \sim 10^{-9}\,\mathrm{m}, \tag{21}$$

$$\delta \sim 0.3\, \delta_{\max}, \tag{22}$$

for some effective domain size $N_{\mathrm{eff}} \ll N$. In this regime

$$x = \frac{\delta}{\delta_{\max}} \sim 0.3, \qquad R(x) \approx \frac{\mathrm{artanh}(0.3)}{0.3} \approx 1.05, \tag{23}$$

so the hyperbolic correction is modest: a few-percent enhancement in τ relative to the linear estimate for the same $(m_{\mathrm{eff}}, \delta)$. Pushing x closer to 1 (near-domain separations) can raise R into the $\mathcal{O}(1.5–3)$ range, but not by orders of magnitude unless one engineers extremely fine-tuned boundary conditions.

The main conceptual impact, therefore, is not a dramatic numerical rescue of Orch-OR, but a principled route to:

- Treat partial separations inside a finite domain with a consistent, associative composition law.
- Interpolate smoothly between tiny displacements (where linearity holds) and near-boundary separations (where the OR-time can be substantially stretched).
- Embed these choices inside a more general information-geometric structure (the MRE/UHL) that also underlies other bounded observables in physics.

4 Spin-Enhanced Coherent Domains: A NiI_2-Inspired Platform

To turn the hyperbolic refinement into a falsifiable physical proposal, we now sketch an experimental platform based on electrically switchable spiral magnetism in layered halides coupled to microtubule-like structures.

4.1 Spiral magnetism and electric control

Nickel(II) iodide (NiI_2) is a prototypical layered halide exhibiting helimagnetic order: below a critical temperature it forms a non-collinear spiral spin texture in which spins rotate from layer to layer with a well-defined pitch [4]. In certain multiferroic materials of this class, the spiral chirality and associated ferroelectric polarization can be switched by applying an external electric field, effectively toggling between left- and right-handed spiral domains.

From the perspective of our framework, such a spiral domain is a candidate for a *spin-coherent region* with:

- A tunable effective mass m_{eff}, set by the number of participating unit cells.
- A tunable domain size $\delta_{\max}$, set by the spatial extent of the spiral and the relevant spin wave mode.
- An externally controllable "internal coordinate" (spiral phase), which can be modulated via electric fields without large dissipation.

4.2 Hybrid microtubule–spin structures

We propose a hybrid device in which:

1. A microtubule or microtubule-mimetic polymer is functionalized or brought into near-field contact with a layered spin material (e.g. a NiI_2-like helimagnet).
2. THz-frequency electromagnetic driving is used both to excite spiral spin modes and to probe putative microtubule conformational superpositions.
3. The spiral chirality is switched in a controlled, time-resolved manner by applied electric fields, modulating the local magnetic field configuration seen by the microtubule structure.

The basic idea is that the spin domain provides a tunable coherent "environment" that can in principle:

- Modify decoherence rates for microtubule conformational states.

- Change the effective mass-displacement profile entering m_{eff} and $\delta_{\max}$.
- Allow the experimentalist to explore regimes in which the dimensionless ratio $x = \delta/\delta_{\max}$ and thus the hyperbolic factor $R(x)$ can be systematically varied.

We emphasize that at present this is a *purely theoretical* proposal. Achieving robust coupling between biological microtubules and cryogenic helimagnets poses significant materials and interface challenges. Nonetheless, even a fully synthetic analog (e.g. a microtubule-like dielectric waveguide coupled to a layered spiral magnet) would already provide a valuable testbed.

4.3 Observable signatures

In such a device, one can search for signatures of hyperbolic partial-separation scaling in:

- Changes in decoherence rates or coherence times of specific vibrational/rotational modes as the spiral domain is tuned.
- Possible correlations between low-frequency (kHz–MHz) signals (analogous to EEG-like activity in bio-inspired networks) and high-frequency (THz) spectroscopy of the hybrid structure, as the effective coherent domain geometry is modulated.

If no such dependencies are observed across a wide, well-characterized parameter sweep, then a broad class of models in which gravitational OR-times are significantly modulated by bounded-geometry effects in such hybrid domains would be strongly constrained.

5 Relation to the Universal Hyperbolic Law / Murray Reality Equation

The hyperbolic map $\chi(\delta)$ in Eq. (11) is not an ad hoc choice. It is the direct analog, for bounded displacement coordinates, of the more general "universal hyperbolic law" proposed by one of us (TM) as a unifying structure for bounded physical observables.

In that broader framework:

- Any physical quantity X that is confined to an interval $(X_{\min}, X_{\max})$ and composes via a continuous, strictly monotone, associative law is conjugate to addition by a map of the form
$$F(X) = \alpha \operatorname{artanh}\left(\frac{2X - X_{\max} - X_{\min}}{X_{\max} - X_{\min}}\right) + \beta.$$
- The "flat coordinate" $F(X)$ plays the role of a rapidity; additive laws in F translate into non-linear but associative laws in X.

Viewed through this lens, our present construction is simply the special case in which X is the displacement parameter inside a coherent domain. The same mathematics that underpins relativistic velocity addition and certain bounded-probability update rules reappears naturally in the gravitational OR context once partial separations are treated as points in a bounded configuration space.

Whether this deeper structural unity is more than a mathematical curiosity depends entirely on experiment. The proposed NiI_2-inspired platform is one way to confront that question.

6 Discussion and Outlook

We summarize our main points:

1. We derived a hyperbolic "partial-separation" map $\chi(\delta)$ from simple axioms: continuity, monotonicity, associativity, and boundedness for a composition law on a finite interval. This yields a refined gravitational self-energy scaling $E_G^{(\chi)} \sim Gm_{\text{eff}}^2/\chi(\delta)$ and corresponding OR-time $\tau_\chi \sim \hbar\chi(\delta)/Gm_{\text{eff}}^2$.

2. The refinement is modest but principled: it recovers the standard linear estimate in the small-displacement regime and introduces a controlled enhancement of τ as separations approach the coherent-domain boundary. It does *not* by itself produce arbitrarily large boosts without fine-tuned geometry.

3. We proposed a concrete, NiI_2-inspired hybrid platform in which electrically switchable spiral magnetism provides tunable coherent domains coupled to microtubule-like structures. Such a system could, in principle, probe whether bounded-geometry effects of the kind encoded in $\chi(\delta)$ play any role in real physical collapse processes.

4. The same hyperbolic structure appears in a broader information-geometric framework (MRE/UHL) that aims to unify diverse bounded observables. The gravitational OR context is thus one of several arenas where this structure may be testable.

From the standpoint of consciousness studies, the present work should be viewed as a *refinement and testing scaffold*, not as a claim that quantum gravity necessarily underlies consciousness. The right standard is not rhetorical plausibility but empirical falsifiability. Our hope is that by making both the mathematics and the experimental knobs explicit, this framework will invite critical engagement from both quantum foundations and quantum biology communities.

Acknowledgements

We thank S. Hameroff and other participants in recent online discussions for clarifying comments about partial separations and microtubule coherence, and the broader community exploring spiral magnetism and multiferroics for inspiration regarding tunable coherent domains. Any errors or speculative overreach are entirely our own.

References

[1] S. Hameroff and R. Penrose, "Consciousness in the universe: A review of the 'Orch OR' theory," *Physics of Life Reviews* **11**, 39–78 (2014).

[2] R. Penrose, "On gravity's role in quantum state reduction," *General Relativity and Gravitation* **28**, 581–600 (1996).

[3] M. Bild *et al.*, "Schrödinger cat states of a 16-microgram mechanical oscillator," *Science* **380**, 274–278 (2023).

[4] See e.g. discussions of helimagnetism and spiral order in layered NiI_2 in: J. X. Yin *et al.*, "Spiral magnetism in layered transition metal halides," (representative review / article; precise citation to be inserted).

[5] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, New York (1966).